Mathematics II

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You can find the last version of these course materials at

https://mbujosab.github.io/MatematicasII/

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Part II
Elementary transformations and elimination

LECTURE 4: Elimination

Lecture 4

(**Lecture** 4) S-1 Highlights of Lesson 4

Highlights of *Lesson 4*

- Elementary transformations
- Identifying singular matrices by elimination
- Matrix multiplication of elementary matrices

F1

Resumen de la lección En esta lección se presenta la mecánica del método de eliminación[1] sin ningún propósito concreto (en la próxima lección se empleará para invertir una matriz; dos lecciones más tarde, para resolver sistemas de ecuaciones; y en la segunda parte del curso, para calcular determinantes y diagonalizar matrices).

La eliminación solo emplea dos tipos de operaciones, denominadas *transformaciones elementales*:

Transformaciones Elementales Tipo I. Suman a un vector un múltiplo de *otro vector*.

Transformaciones Elementales Tipo II. Multiplican un vector por un número *distinto de cero*.

Una vez introducidos los dos tipos de transformaciones elementales, se propone pre-escalonar una matriz de tres por tres. Con este ejercicio se ilustra en qué consiste una *matriz (pre)escalonada* y a qué denominamos *pivote*: el pivote de una columna no nula es su primer componente no nulo. Una matriz está *pre-escalonada* si los componentes a la derecha de cada pivote son nulos, y está *escalonada* si, además, los pivotes de sus columnas describen una escalera descendente, dejando todas las columnas nulas (si las hay) a la derecha de las columnas no nulas.

Posteriormente se propone a los estudiantes que intenten pre-escalonar varias matrices de orden 3 y que comprueben si la forma pre-escalonada tiene tres pivotes (o si, por el contrario, tiene columnas nulas). La idea es darles unos minutos para que lo intenten y comprendan la mecánica (a lo largo del curso casi todo se resolverá mediante eliminación, así que es fundamental comprender la mecánica cuanto antes).

Después se mostrará cómo realizar una transformación elemental $\boldsymbol{\tau}$ de las columnas de $\mathbf{A}$ mediante un producto de matrices $\mathbf{A}\big(\mathbf{I}_{\boldsymbol{\tau}}\big)$, donde $\big(\mathbf{I}_{\boldsymbol{\tau}}\big)$ es la *matriz elemental* resultante de aplicar una sola transformación elemental $\boldsymbol{\tau}$ a las columnas de la matriz identidad. Es importante explicar la notación usada para denotar las transformaciones elementales de la columna j-ésima (que también describen las correspondientes matrices elementales): $\mathbf{I}_{\underset{[(a)\boldsymbol{k}+\boldsymbol{j}]}{\boldsymbol{\tau}}}$ y $\mathbf{I}_{\underset{[(a)\boldsymbol{j}]}{\boldsymbol{\tau}}}$.

Seguidamente veremos una de las ideas que más usaremos durante el curso: que una sucesión de transformaciones elementales de las columnas de $\mathbf{A}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}$ se puede expresar como el producto de $\mathbf{A}$ por una matriz que es el producto de todas las matrices elementales empleadas en la sucesión: $\mathbf{I}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k} = \mathbf{I}_{\boldsymbol{\tau}_1}\cdots\mathbf{I}_{\boldsymbol{\tau}_k}$.

$$\mathbf{A}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k} = \mathbf{A}\big(\mathbf{I}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}\big)$$

Esta idea es el pilar del método de eliminación.

Después destacaremos que las transformaciones elementales se pueden *deshacer*, es decir, que son *invertibles* (esta es una propiedad fundamental y será central en la lección sobre matrices invertibles).

También veremos que el intercambio de posición de dos vectores resulta ser equivalente a una sucesión de transformaciones *Tipo I* y *II* —se debe mostrar en la pizarra con la matriz identidad de orden 2, animando a los estudiantes a que indiquen los pasos necesarios. Es interesante que vean que hay varias formas de llegar al mismo resultado. Llamaremos *permutación*, $\underset{[\mathfrak{S}]}{\boldsymbol{\tau}}$, a cualquier sucesión arbitraria de intercambios; consecuentemente, $\mathbf{I}_{\underset{[\mathfrak{S}]}{\boldsymbol{\tau}}}$ es una reordenación de las columnas de $\mathbf{I}$. Veremos que, puesto que $\underset{[\mathfrak{S}]}{\boldsymbol{\tau}}$ es una sucesión de intercambios, $\mathbf{A}\big(\mathbf{I}_{\underset{[\mathfrak{S}]}{\boldsymbol{\tau}}}\big)$ reordena las columnas de $\mathbf{A}$.

[1] En esta lección solo aplicaremos la eliminación "de izquierda a derecha".

La demostración de que toda matriz se puede escalonar mediante transformaciones elementales no se cuenta en clase, pero dicha demostración es básicamente una descripción del método de eliminación (véase el libro y la documentación de la librería de Python que lo acompaña).

(**Lecture** 4) **S-2** **Elementary transformations of a matrix**

Type *I*: $\mathbf{A}_{\underset{[(\lambda)\boldsymbol{i}+\boldsymbol{j}]}{\boldsymbol{\tau}}}$ ***(with*** $i \neq j$***)***

add λ times i-th column ($\lambda\mathbf{A}_{|i}$) to j-th column ($\mathbf{A}_{|j}$)

$$\begin{bmatrix} 1 & -3 & 0 \\ 1 & -6 & 3 \end{bmatrix}_{\underset{[(-2)\boldsymbol{1}+\boldsymbol{3}]}{\boldsymbol{\tau}}} = \begin{bmatrix} 1 & -3 & -2 \\ 1 & -6 & 1 \end{bmatrix}$$

Type *II*: $\mathbf{A}_{\underset{[(\alpha)\boldsymbol{i}]}{\boldsymbol{\tau}}}$ ***(with*** $\alpha \neq 0$***)***

multiply by α the i-th column

$$\begin{bmatrix} 1 & -3 & 0 \\ 1 & -6 & 3 \end{bmatrix}_{\underset{[(10)\boldsymbol{2}]}{\boldsymbol{\tau}}} = \begin{bmatrix} 1 & -30 & 0 \\ 1 & -60 & 3 \end{bmatrix}$$

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Type I transformation of the identity matrix of order 4:

$$\mathbf{I}_{\underset{[(-3)\boldsymbol{2}+\boldsymbol{4}]}{\boldsymbol{\tau}}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Type II transformation of the identity matrix of order 4:

$$\mathbf{I}_{\underset{[(5)\boldsymbol{3}]}{\boldsymbol{\tau}}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Pivots and echelon form of a matrix $\mathbf{A}$: We call the *pivot* of a *non-zero* column the first non-zero component of that column; and we refer to the *pivot position* as the index of the row where that *non-zero* coefficient is located.

We say a matrix $\mathbf{K}$ is *pre-echelon* if the components to the right of each pivot are zero. We say a matrix $\mathbf{L}$ is *echelon* if it is pre-echelon, and every column that precedes a non-zero column $\mathbf{A}_{|k}$ is non-zero and has a pivot position earlier than that of $\mathbf{A}_{|k}$.

Therefore, in an echelon matrix, there cannot be zero columns to the left of non-zero columns (meaning there are either no zero columns, or all are located on the right side of the matrix). Moreover, as we move from left to right, the pivots appear further down (i.e., in rows with progressively larger indices). Thus, traversing the matrix from left to right, the pivots describe a descending staircase (hence the name "echelon" matrix).

(**Lecture** 4) **S-3** **Elimination and pre-echelon form of a matrix**

- *Pivot* is the first non-zero component of each column.
- *Elimination*: modifies a matrix until all *components at the right-hand side of each pivot are zeros*

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & 0 \\ 2 & 8 & 4 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow{\underset{[(-3)\boldsymbol{1}+\boldsymbol{2}]}{\boldsymbol{\tau}}} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 4 \\ 1 & -2 & 1 \end{bmatrix} \xrightarrow{\underset{[(-2)\boldsymbol{2}+\boldsymbol{3}]}{\boldsymbol{\tau}}} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 1 & -2 & 5 \end{bmatrix} = \mathbf{L}$$

F4

(**Lecture** 4) **S-4** **Elimination**

Elimination algorithm on $\mathbf{A}$
modifies $\mathbf{A}$ using a sequence of *elementary transformations*

Goal
to get a *(pre)echelon* form

- *pre-echelon*: all components on the right side of each pivot are zero.
- *echelon*: if any column before a non-null column $\mathbf{A}_{|j}$ is non-null and its pivot is above the pivot of $\mathbf{A}_{|j}$.

It is always possible to find a (pre)echelon form by elimination
Rank (rg): the number of pivots in any of its pre-echelon forms
$\underset{n\times n}{\mathbf{A}}$ is *singular* if its pre-echelon forms have null-columns (rg $< n$)

F5

(**Lecture** 4) **S-5** **Elimination: When can't we find n pivots?**

$n \times n$ matrices are **singular** if less than n pivots after elimination

$$\begin{bmatrix} 0 & 1 & 3 \\ 4 & 2 & 8 \\ 1 & 1 & 1 \end{bmatrix}$$

Has this matrix n pivots? $\begin{bmatrix} 1 & 3 & 0 \\ 2 & 6 & 4 \\ 1 & 1 & 1 \end{bmatrix}$

and this one? $\begin{bmatrix} 1 & 3 & 0 \\ 2 & 6 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

and this one? $\begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & -4 \end{bmatrix}$

F6

(**Lecture** 4) **S-6** **Matrix multiplication: elementary matrices**

$$\underbrace{\begin{bmatrix} 1 & 3 & 0 \\ 2 & 8 & 4 \\ 1 & 1 & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} \quad & \quad & \quad \\ \quad & \quad & \quad \\ \quad & \quad & \quad \end{bmatrix}}_{\mathbf{I}_\tau} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 4 \\ 1 & -2 & 1 \end{bmatrix}}_{\mathbf{A}_\tau}$$

We call $\mathbf{I}_\tau$ "Elementary matrix":

$$\mathbf{A}\left(\mathbf{I}_\tau\right) = \mathbf{A}_\tau$$

This specific elementary matrix $\mathbf{I}_\tau$ is written as $\mathbf{I}_{\underset{[(-3)\mathbf{1}+\mathbf{2}]}{\tau}}$

$$\mathbf{A}\left(\mathbf{I}_{\underset{[(-3)\mathbf{1}+\mathbf{2}]}{\tau}}\right) = \mathbf{A}_{\underset{[(-3)\mathbf{1}+\mathbf{2}]}{\tau}}$$

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(**Lecture** 4) **S-7** **Matrix multiplication: elementary matrices**

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 4 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} \quad & \quad & \quad \\ \quad & \quad & \quad \\ \quad & \quad & \quad \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 1 & -2 & 5 \end{bmatrix}$$

This specific elementary matrix $\mathbf{I}_{\tau}$ is written as $\mathbf{I}_{\substack{\tau \\ [(-2)\mathbf{2}+\mathbf{3}]}}$

$$\mathbf{A}\left(\mathbf{I}_{\substack{\tau \\ [(-2)\mathbf{2}+\mathbf{3}]}}\right) = \mathbf{A}_{\substack{\tau \\ [(-2)\mathbf{2}+\mathbf{3}]}}$$

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(**Lecture** 4) **S-8** **Elimination by elementary matrices**

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & 0 \\ 2 & 8 & 4 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow{[(-3)\overset{\tau}{\mathbf{1}+\mathbf{2}}]} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 4 \\ 1 & -2 & 1 \end{bmatrix} \xrightarrow{[(-2)\overset{\tau}{\mathbf{2}+\mathbf{3}}]} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 1 & -2 & 5 \end{bmatrix} = \mathbf{L}$$

$$\mathbf{A}_{\substack{\tau \\ [(-3)\mathbf{1}+\mathbf{2}] \\ [(-2)\mathbf{2}+\mathbf{3}]}} = \mathbf{A}_{\substack{\tau \\ [(-3)\mathbf{1}+\mathbf{2}]}\substack{\tau \\ [(-2)\mathbf{2}+\mathbf{3}]}} = \left(\mathbf{A}\left(\mathbf{I}_{\substack{\tau \\ [(-3)\mathbf{1}+\mathbf{2}]}}\right)\right)\left(\mathbf{I}_{\substack{\tau \\ [(-2)\mathbf{2}+\mathbf{3}]}}\right) = \mathbf{L}$$

there is a matrix that does the whole job *"at once"*

$$\mathbf{A}_{\substack{\tau \\ [(-3)\mathbf{1}+\mathbf{2}] \\ [(-2)\mathbf{2}+\mathbf{3}]}} = \mathbf{A}\left(\left(\mathbf{I}_{\substack{\tau \\ [(-3)\mathbf{1}+\mathbf{2}]}}\right)\left(\mathbf{I}_{\substack{\tau \\ [(-2)\mathbf{2}+\mathbf{3}]}}\right)\right) = \mathbf{A}\mathbf{I}_{\substack{\tau \\ [(-3)\mathbf{1}+\mathbf{2}] \\ [(-2)\mathbf{2}+\mathbf{3}]}} = \mathbf{L}$$

$$\boxed{\mathbf{A}_{\tau_1 \cdots \tau_k} = \mathbf{A}\left(\mathbf{I}_{\tau_1 \cdots \tau_k}\right)}$$

F9

(**Lecture** 4) **S-9** **How do I get from L back to A? Inverses**

How do I reverse the first step? (it was subtract 3 times $\mathbf{A}_{|1}$ from $\mathbf{A}_{|2}$)

$$\begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \quad & \quad & \quad \\ \quad & \quad & \quad \\ \quad & \quad & \quad \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$\mathbf{I}_{\substack{\tau \\ [(-\lambda)\boldsymbol{i}+\boldsymbol{j}]}} \quad \text{"undo"} \quad \mathbf{I}_{\substack{\tau \\ [(\lambda)\boldsymbol{i}+\boldsymbol{j}]}}$$

How to undo $\mathbf{I}_{\substack{\tau \\ [(\alpha)\boldsymbol{i}]}}$?

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} \quad & \quad & \quad \\ \quad & \quad & \quad \\ \quad & \quad & \quad \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

F10

(**Lecture** 4) **S-10** **Interchange or swap matrices**

Which matrix exchanges the columns?

$$\begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix} = \begin{bmatrix} c & a \\ d & b \end{bmatrix}$$

Which matrix exchanges the rows? where do we put that matrix?

$$\begin{bmatrix} a & c \\ b & d \end{bmatrix} \qquad = \begin{bmatrix} b & d \\ a & c \end{bmatrix}$$

Matrix multiplication is not commutative!

F11

(**Lecture** 4) **S-11** **Interchange of columns**

Interchange of columns:

$\mathbf{A}_{\underset{[i \rightleftharpoons j]}{\tau}} \rightarrow$ switch columns i and j of $\mathbf{A}$

$$\begin{bmatrix} 1 & -3 & 0 \\ 1 & -6 & 3 \end{bmatrix}_{\underset{[\mathbf{2} \rightleftharpoons \mathbf{3}]}{\tau}} = \begin{bmatrix} 1 & 0 & -3 \\ 1 & 3 & -6 \end{bmatrix}$$

We can switch two columns by a sequence of elementary transformations

Matrix $\mathbf{I}_{\underset{[i \rightleftharpoons j]}{\tau}}$ is called an exchange matrix

F12

(**Lecture** 4) **S-12** **Permutation matrices**

Product between exchange matrices $\mathbf{I}_{\underset{[\text{-} \rightleftharpoons \text{-}]}{\tau}}$ is a permutation matrix $\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}$.

$$\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}} = \text{Identity matrix } \mathbf{I} \text{ with rearranged columns}$$

Let's see the 3×3 case

$$\begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}, \quad \mathbf{I}_{\underset{[\mathbf{1} \rightleftharpoons \mathbf{2}]}{\tau}} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

How many 3×3 permutations can we find?

What happens if I multiply two permutation matrices?

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The lecture ends here

Questions of the Lecture 4

(L-4) Question 1.

(a) What are the matrices $\mathbf{I}_{\underset{[(x)\mathbf{1}+\mathbf{2}]}{\tau}}$, $\mathbf{I}_{\underset{[(y)\mathbf{1}+\mathbf{3}]}{\tau}}$ and $\mathbf{I}_{\underset{[(z)\mathbf{2}+\mathbf{3}]}{\tau}}$ that transform $\mathbf{A} = \begin{bmatrix} 1 & 4 & -2 \\ 1 & 6 & 2 \\ 0 & 1 & 0 \end{bmatrix}$ into echelon form?

(b) Multiply those $\mathbf{I}_{\tau_i}$ to obtain a matrix $\mathbf{E}$ that performs the elimination: $\mathbf{AE} = \mathbf{K}$.

Based on (Strang, 1988, exercise 24 from section 1.4.)

(L-4) QUESTION 2. Consider the matrix

$$\begin{bmatrix} 1 & 2 & 4 \\ -1 & -3 & -2 \\ 0 & 1 & c \end{bmatrix}$$

For what value(s) of c is the matrix singular (meaning that we cannot find three pivots)?

(L-4) QUESTION 3. Consider the following 3 by 3 matrices.

(a) $\left(\mathbf{I}_{\substack{\tau \\ [(-1)\mathbf{1}+\mathbf{2}]}}\right)$ subtracts column 1 from column 2 and then $\left(\mathbf{I}_{\substack{\tau \\ [\mathbf{2}\rightleftharpoons\mathbf{3}]}}\right)$ exchanges columns 2 and 3. What matrix $\mathbf{E}$ performs both changes at once?

(b) $\left(\mathbf{I}_{\substack{\tau \\ [\mathbf{2}\rightleftharpoons\mathbf{3}]}}\right)$ exchanges columns 2 and 3 and then $\mathbf{I}_{\substack{\tau \\ [(-1)\mathbf{1}+\mathbf{3}]}}$ subtracts column 1 from column 3. What matrix $\mathbf{N} = \left(\mathbf{I}_{\substack{\tau \\ [\mathbf{2}\rightleftharpoons\mathbf{3}]}}\right)\left(\mathbf{I}_{\substack{\tau \\ [(-1)\mathbf{1}+\mathbf{3}]}}\right)$ performs both changes at once? Explain why $\mathbf{E}$ and $\mathbf{N}$ are the same in both cases, while the elementary matrices $\mathbf{I}_{\tau}$ are different.

Based on (Strang, 1988, exercise 28 from section 1.4.)

(L-4) QUESTION 4. The elementary matrices $\mathbf{I}_{\substack{\tau \\ [(?)\mathbf{1}+\mathbf{2}]}}$ and $\mathbf{I}_{\substack{\tau \\ [(?)\mathbf{2}+\mathbf{3}]}}$ reduce $\mathbf{A}$ to its echelon form. Find $\mathbf{E}$ so that $\mathbf{AE} = \mathbf{L}$ is an echelon form (lower triangular), if $\mathbf{A}$ is

$$\begin{bmatrix} 2 & 2 & 0 \\ 1 & 4 & 9 \\ 1 & 3 & 9 \end{bmatrix}$$

(L-4) QUESTION 5. Although we will only consider the *Type I* and *II* elementary transformations here, most Linear Algebra textbooks include a third type: the *exchange* of columns

$$\mathbf{A}_{\substack{\tau \\ [p\rightleftharpoons s]}} \rightarrow \text{Exchanges columns } p \text{ and } s \text{ of } \mathbf{A}.$$

Prove that a column exchange is, in fact, a sequence of *Type I* and *II* elementary transformations. Do this by transforming the identity matrix $\mathbf{I}_{2\times 2}$ into $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ using column elementary transformations.

(L-4) QUESTION 6. Write the 3 by 3 matrices that produce the following elimination steps:

(a) $\mathbf{I}_{\substack{\tau \\ [(-5)\mathbf{1}+\mathbf{2}]}}$ subtracts 5 times column 1 from column 2,

(b) $\mathbf{I}_{\substack{\tau \\ [(-7)\mathbf{2}+\mathbf{3}]}}$ subtracts 7 times column 2 from column 3,

(c) $\mathbf{I}_{\substack{\tau \\ [\circlearrowleft]}}$ exchanges columns 1 and 2, and then columns 2 and 3.

(Strang, 2003, exercise 1 from section 2.3.)

(L-4) QUESTION 7. Consider the matrices from PROBLEM 6:

(a) when multiplying by $\mathbf{I}_{\substack{\tau \\ [(-5)\mathbf{1}+\mathbf{2}]}}$ and then by $\mathbf{I}_{\substack{\tau \\ [(-7)\mathbf{2}+\mathbf{3}]}}$ the matrix $\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$ we obtain $\mathbf{A}_{\substack{\tau \\ [(-5)\mathbf{1}+\mathbf{2}] \\ [(-7)\mathbf{2}+\mathbf{3}]}} = \begin{bmatrix} \quad ; & \quad ; & \quad \end{bmatrix}$.

(b) But, when multiplying by $\mathbf{I}_{\substack{\tau \\ [(-5)\mathbf{1}+\mathbf{2}]}}$ before and then by $\mathbf{I}_{\substack{\tau \\ [(-7)\mathbf{2}+\mathbf{3}]}}$ we get $\mathbf{A}_{\substack{\tau \\ [(-7)\mathbf{2}+\mathbf{3}] \\ [(-5)\mathbf{1}+\mathbf{2}]}} = \begin{bmatrix} \quad ; & \quad ; & \quad \end{bmatrix}$.

(c) When $\mathbf{I}_{\substack{\tau \\ [(-7)\mathbf{2}+\mathbf{3}]}}$ comes first, the column ____ is not affected by column ____. This fact is critical for how the LU factorization works!

(Strang, 2003, exercise 2 from section 2.3.)

(L-4) QUESTION 8. What matrix $\mathbf{M}$ transforms the vector $\boldsymbol{v} = \begin{pmatrix} 1, & 0, \end{pmatrix}$ to $\begin{pmatrix} 0, & 1, \end{pmatrix}$, es decir $\boldsymbol{v}\mathbf{M} = \begin{pmatrix} 0, & 1, \end{pmatrix}$; and also transforms the vector $\boldsymbol{w} = \begin{pmatrix} 0, & 1, \end{pmatrix}$ to $\begin{pmatrix} 1, & 0, \end{pmatrix}$, es decir $\boldsymbol{w}\mathbf{M} = \begin{pmatrix} 1, & 0, \end{pmatrix}$?

(L-4) Question 9. Consider an interchange matrix $\mathbf{I}_{\underset{[i \rightleftharpoons j]}{\tau}}$, if we compute the product $\mathbf{A}\left(\mathbf{I}_{\underset{[i \rightleftharpoons j]}{\tau}}\right)$, we get a new matrix like $\mathbf{A}$, but with exchanged columns. What happen if we compute the product $\left(\mathbf{I}_{\underset{[i \rightleftharpoons j]}{\tau}}\right)\mathbf{A}$? Check your answer with a 2 by 2 example.

(L-4) Question 10. If each column of $\mathbf{A}$ is a multiple of $\begin{pmatrix}1, & 1, & 1,\end{pmatrix}$, then $\mathbf{A}\boldsymbol{x}$ is always a multiple of $\begin{pmatrix}1, & 1, & 1,\end{pmatrix}$. Write a 3 by 3 example. How many pivots are produced by elimination?
(Strang, 1988, exercise 26 from section 1.4.)

End of Questions of the Lecture 4

LECTURE 5: Inverse matrices

Lecture 5

(**Lecture** 5) S-1 Highlights of Lesson 5

Highlights of *Lesson 5*

- Inverse of $\mathbf{A}$
- Gauss-Jordan elimination / finding $\mathbf{A}^{-1}$
- Inverse of $\mathbf{AB}$, $\mathbf{A}^{\intercal}$

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Resumen de la lección: La inversa de una matriz cuadrada $\mathbf{A}$ es una matriz $\mathbf{B}$ tal que

$$\mathbf{AB} = \mathbf{BA} = \mathbf{I}.$$

Es importante indicar que no toda matriz cuadrada tiene inversa... y expondremos dos argumentos: por una parte, es imposible combinar las columnas de $\mathbf{A}$ para generar las columnas de $\mathbf{I}$ si las columnas de $\mathbf{A}$ están alineadas. Por otra, y de forma más general, es imposible que se puedan combinar las columnas de $\mathbf{A}$ de manera que $\mathbf{A}\boldsymbol{x} = \mathbf{0}$, con $\boldsymbol{x} \neq \mathbf{0}$, y que al mismo tiempo exista $\mathbf{A}^{-1}$.

Veremos que encontrar la inversa de $\mathbf{A}$ de orden n implica resolver n sistemas de ecuaciones con n incógnitas (pues hemos de encontrar n combinaciones de las columnas de $\mathbf{A}$ que generen las n columnas de $\mathbf{I}$, una combinación por cada columna de $\mathbf{I}$).

La aparente "magia" del método de Gauss consiste en que es posible resolver simultáneamente los n sistemas de ecuaciones. Para hacerlo basta alargar las columnas de $\mathbf{A}$ con las columnas de la matriz identidad y aplicar después la eliminación Gauss-Jordan; para entenderlo hay que recordar que aplicar una secuencia de transformaciones elementales es equivalente a multiplicar la matriz por la matriz resultante de aplicar dicha sucesión sobre la matriz identidad:

$$\mathbf{A}_{\tau_1 \cdots \tau_k} = \mathbf{A}\left(\mathbf{I}_{\tau_1 \cdots \tau_k}\right)$$

También deduciremos cómo es la inversa tanto del producto de matrices invertibles, $(\mathbf{AB})^{-1}$, como de la transpuesta de una matriz invertible, $(\mathbf{A}^{\intercal})^{-1}$.

Para finalizar, daremos una caracterización de las matrices invertibles relacionada con el método de eliminación y las matrices elementales (la demostración formal aparece en el libro, pero tras haber aplicado Gauss-Jordan con un ejemplo en la pizarra, se puede explicar la idea que hay detrás de esta caracterización).

(**Lecture** 5) S-2 Inverse of a matrix (square matrices)

$\mathbf{A}$ squared of order n has an inverse (is *invertible*) if it exists $\mathbf{B}$ such that

$$\mathbf{AB} = \mathbf{BA} = \mathbf{I}.$$

Then

$$\mathbf{B} = \mathbf{A}^{-1} \qquad \text{and} \qquad \mathbf{A} = \mathbf{B}^{-1}.$$

Not all matrices have inverses

Square matrices with no inverses are called *singular* matrices

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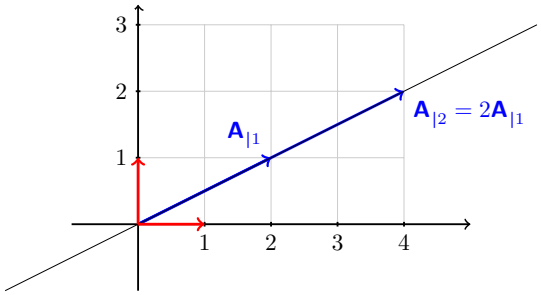
(**Lecture** 5) S-3 **Singular case (no inverse)**

$$\mathbf{A} = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$$

Is it possible to find a matrix $\mathbf{B}$ such that $\mathbf{AB} = \mathbf{I}$?
...columns of $\mathbf{I}$ must be linear combinations of columns of $\mathbf{A}$...but the columns are aligned.



So

$$\boxed{\mathbf{A} \text{ is singular}}$$

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(**Lecture** 5) S-4 **Singular case (no inverse)**

Can we find $\boldsymbol{x} \neq \mathbf{0}$ such that $\mathbf{A}\boldsymbol{x} = \mathbf{0}$?

$$\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \begin{pmatrix} \quad \\ \quad \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

If $\mathbf{A}\boldsymbol{x} = \mathbf{0}$ and $\boldsymbol{x} \neq \mathbf{0}$ $\Rightarrow$ there is no $\mathbf{A}^{-1}$

$$\boxed{\text{The existence of } \mathbf{A}^{-1} \text{ leads to a } \textit{contradiction}}$$

$$\text{If } \mathbf{A}\boldsymbol{x} = \mathbf{0} \text{ and } \boldsymbol{x} \neq \mathbf{0} \quad \Rightarrow \quad \mathbf{A}^{-1}\mathbf{A}\boldsymbol{x} = \mathbf{A}^{-1}\mathbf{0} \quad \Rightarrow \quad \boldsymbol{x} = \mathbf{0}.$$

When $\mathbf{A}^{-1}$ does exist

the only solution to $\mathbf{A}\boldsymbol{x} = \mathbf{0}$ is $\boldsymbol{x} = \mathbf{0}$.

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(**Lecture** 5) S-5 **Calculating the inverse matrix**

$$\mathbf{A}(\mathbf{A}^{-1}) = \mathbf{I}$$

$$\begin{bmatrix} 1 & 3 \\ 2 & \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

So...we are solving m systems (of m equations each)

$$\begin{bmatrix} 1 & 3 \\ 2 & \end{bmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \begin{bmatrix} 1 & 3 \\ 2 & \end{bmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

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(**Lecture** 5) **S-6** **Gauss-Jordan: solving two linear systems at once**

Gauss-Jordan elimination (obtaining a reduced echelon form R)
apply elementary transformations until a echelon matrix with only zeros to the left of each pivot (and all pivots equal to 1) is achieved

Let's solve the linear systems

$$\begin{bmatrix}1 & 3\\ 2 & 7\end{bmatrix}\begin{pmatrix}a\\ b\end{pmatrix}=\begin{pmatrix}1\\ 0\end{pmatrix} \qquad \begin{bmatrix}1 & 3\\ 2 & 7\end{bmatrix}\begin{pmatrix}c\\ d\end{pmatrix}=\begin{pmatrix}0\\ 1\end{pmatrix}$$

applying Gauss-Jordan elimination on $\mathbf{A}$ stacked with $\mathbf{I}$

$$\left[\begin{array}{c}\mathbf{A}\\ \hline \mathbf{I}\end{array}\right]=\left[\begin{array}{cc}1 & 3\\ 2 & 7\\ \hline 1 & 0\\ 0 & 1\end{array}\right] \rightarrow \qquad \rightarrow \qquad =$$

If $\mathbf{R}=\mathbf{I}$, we have found $\mathbf{A}^{-1}$

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```
A = Matrix([[1,3],[2,7]])
B = ElimGJ( A.apila(I(2),1) , 1 )
(3,4)|B
```

(**Lecture** 5) **S-7** **Gauss-Jordan: Why does it work?**

$$\left[\begin{array}{c}\mathbf{A}\\ \hline \mathbf{I}\end{array}\right]=\left[\begin{array}{cc}1 & 3\\ 2 & 7\\ \hline 1 & 0\\ 0 & 1\end{array}\right] \xrightarrow[{[(-3)\mathbf{1}+\mathbf{2}]}]{\boldsymbol{\tau}} \qquad \xrightarrow[{[(-2)\mathbf{2}+\mathbf{1}]}]{\boldsymbol{\tau}}$$

that is, since $\mathbf{A}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}=\mathbf{A}\left(\mathbf{I}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}\right)$:

$$\left[\begin{array}{c}\mathbf{A}\\ \hline \mathbf{I}\end{array}\right]_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}=\left[\begin{array}{c}\mathbf{A}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}\\ \hline \mathbf{I}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}\end{array}\right]=\left[\begin{array}{c}\mathbf{A}(\mathbf{I}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k})\\ \hline \mathbf{I}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}\end{array}\right]=\left[\begin{array}{c}\mathbf{I}\\ \hline \mathbf{I}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}\end{array}\right],$$

$$\boxed{\textit{who is } \mathbf{I}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}\textit{?}}$$

therefore $\mathbf{A}^{-1}=$

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$$\text{Since}\quad \mathbf{I}_{\underset{[(-3)\mathbf{1}+\mathbf{2}]}{\boldsymbol{\tau}}}=\begin{bmatrix}1 & -3\\ 0 & 1\end{bmatrix},\quad \text{and}\quad \mathbf{I}_{\underset{[(-2)\mathbf{2}+\mathbf{1}]}{\boldsymbol{\tau}}}=\begin{bmatrix}1 & 0\\ -2 & 1\end{bmatrix};$$

then

$$\mathbf{I}_{\underset{[(-3)\mathbf{1}+\mathbf{2}]}{\boldsymbol{\tau}}\underset{[(-2)\mathbf{2}+\mathbf{1}]}{\boldsymbol{\tau}}}=\left(\mathbf{I}_{\underset{[(-3)\mathbf{1}+\mathbf{2}]}{\boldsymbol{\tau}}}\right)\left(\mathbf{I}_{\underset{[(-2)\mathbf{2}+\mathbf{1}]}{\boldsymbol{\tau}}}\right)=\begin{bmatrix}1 & -3\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ -2 & 1\end{bmatrix}=\begin{bmatrix}7 & -3\\ -2 & 1\end{bmatrix}.$$

Since $\mathbf{A}\left(\mathbf{I}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}\right)=\mathbf{I}$, it follows that $\mathbf{I}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}$ is the inverse matrix of $\mathbf{A}$.

$$\boxed{\text{In other words, if } \mathbf{A}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}=\mathbf{I} \text{ then } \mathbf{I}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}=\mathbf{A}^{-1}.}$$

We can also compute the inverse by operating with the rows (the usual method in most textbooks). In this case, we multiply from the left to work with the rows.

$$[\mathbf{A}|\mathbf{I}]=\left[\begin{array}{cc|cc}1 & 3 & 1 & 0\\ 2 & 7 & 0 & 1\end{array}\right]\xrightarrow[{[(-2)\mathbf{1}+\mathbf{2}]}]{\boldsymbol{\tau}}\left[\begin{array}{cc|cc}1 & 3 & 1 & 0\\ 0 & 1 & -2 & 1\end{array}\right]\xrightarrow[{[(-3)\mathbf{2}+\mathbf{1}]}]{\boldsymbol{\tau}}\left[\begin{array}{cc|cc}1 & 0 & 7 & -3\\ 0 & 1 & -2 & 1\end{array}\right]=\left[{}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}\mathbf{A}\,\middle|\,{}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}\mathbf{I}\right]=\left[\mathbf{I}\,\middle|\,{}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}\mathbf{I}\right].$$

What is the matrix ${}_{\tau_1\cdots\tau_k}\mathbf{I}$ on the right-hand side? The reasoning is the same:

$$ {}_{\tau_1\cdots\tau_k}\left[\mathbf{A}\,|\,\mathbf{I}\right] = \left[\left({}_{\tau_1\cdots\tau_k}\mathbf{I}\right)\mathbf{A}\,|\,{}_{\tau_1\cdots\tau_k}\mathbf{I}\right] = \left[\mathbf{I}\,|\,{}_{\tau_1\cdots\tau_k}\mathbf{I}\right] $$

Since $\left({}_{\tau_1\cdots\tau_k}\mathbf{I}\right)\mathbf{A} = \mathbf{I}$, then ${}_{\tau_1\cdots\tau_k}\mathbf{I} = \mathbf{A}^{-1}$.

$$ \boxed{\text{In other words, if } {}_{\tau_1\cdots\tau_k}\mathbf{A} = \mathbf{I} \text{ then } {}_{\tau_1\cdots\tau_k}\mathbf{I} = \mathbf{A}^{-1}.} $$

(**Lecture** 5) **S-8** **Inverse of a product**

If $\mathbf{A}$ and $\mathbf{B}$ are invertible and of the same order, $(\mathbf{AB})$ is invertible.
How do we find the inverse of $\mathbf{AB}$? Let's try using $(\mathbf{B}^{-1}\mathbf{A}^{-1})$:

$$ \mathbf{AB}\left(\mathbf{B}^{-1}\mathbf{A}^{-1}\right) = $$

$$ \left(\mathbf{B}^{-1}\mathbf{A}^{-1}\right)\mathbf{AB} = $$

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(**Lecture** 5) **S-9** **Inverse of a transpose matrix**

$$ \mathbf{AA}^{-1} = \mathbf{I} $$

Let's take the transpose of both sides

$$ \left((\mathbf{A}^{-1})^{\mathsf{T}}\right)\mathbf{A}^{\mathsf{T}} = \mathbf{I} $$

Therefore

$$ \boxed{\text{the inverse of } \mathbf{A}^{\mathsf{T}} \text{ is}} $$

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(**Lecture** 5) **S-10** **Interchange matrices and permutation matrices**

Are interchange matrices $\mathbf{I}_{\underset{[i\rightleftarrows j]}{\tau}}$, invertible?
It is easy to demonstrate that

$$ \left(\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}\right)^{\mathsf{T}}\left(\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}\right) = \mathbf{I} \qquad \Longrightarrow $$

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Exercise 1. Prove that $\left(\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}\right)^{\mathsf{T}}\left(\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}\right) = \mathbf{I}$.

(**Lecture** 5) $\boxed{\text{S-11}}$ **Characterization of invertible matrices**

Given $\mathbf{A}$ of order n, the following statements are equivalent

1. No zero columns in $\mathbf{A}_{\tau_1\cdots\tau_p} = \mathbf{K}$ (pre-echelon matrix).
2. $\mathbf{A}$ is invertible.
3. $\mathbf{A}$ is a product of elementary matrices.

$$\mathbf{A}_{\tau_1\cdots\tau_k} = \mathbf{A}\left(\mathbf{I}_{\tau_1\cdots\tau_k}\right) = \mathbf{I} \qquad \Rightarrow \qquad \mathbf{A} = \left(\mathbf{I}_{\tau_1\cdots\tau_k}\right)^{-1}$$

where

$$\left(\mathbf{I}_{\tau_1\cdots\tau_k}\right)^{-1} = \left((\mathbf{I}_{\tau_1})\cdots(\mathbf{I}_{\tau_k})\right)^{-1} = (\mathbf{I}_{\tau_k^{-1}})\cdots(\mathbf{I}_{\tau_1^{-1}}) = \mathbf{I}_{\tau_k^{-1}\cdots\tau_1^{-1}}$$

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- If some columns of $\underset{n\times n}{\mathbf{A}}$ are aligned (if some are linear combinations of others), the matrix is singular.
- If the system of equations $\mathbf{A}\boldsymbol{x} = \mathbf{0}$ has more than one solution, then the square matrix $\mathbf{A}$ is singular.
- A square matrix $\mathbf{A}$ is singular if any of its pre-echelon forms have null columns.
- Finding the inverse of $\underset{n\times n}{\mathbf{A}}$ involves solving n systems of equations.
- The inverse of $\mathbf{A}$ is the product of the elementary matrices (elementary operations) that transform $\mathbf{A}$ into $\mathbf{I}$:

$$\left[\begin{array}{c}\mathbf{A}\\ \hline \mathbf{I}\end{array}\right]_{\tau_1\cdots\tau_k} = \left[\begin{array}{c}\mathbf{A}(\mathbf{I}_{\tau_1\cdots\tau_k})\\ \hline \mathbf{I}_{\tau_1\cdots\tau_k}\end{array}\right]; \qquad \text{si} \quad \mathbf{A}\left(\mathbf{I}_{\tau_1\cdots\tau_k}\right) = \mathbf{I} \quad \text{entonces} \quad \mathbf{I}_{\tau_1\cdots\tau_k} = \mathbf{A}^{-1}.$$

- $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$
- $(\mathbf{A}^{-1})^{\mathsf{T}} = (\mathbf{A}^{\mathsf{T}})^{-1}$
- $\left(\mathbf{I}_{\underset{[j\rightleftharpoons k]}{\tau}}\right)^{-1} = \mathbf{I}_{\underset{[j\rightleftharpoons k]}{\tau}}$
- $\left(\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}\right)^{\mathsf{T}}\left(\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}\right) = \mathbf{I} \qquad \Rightarrow \qquad \left(\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}\right)^{\mathsf{T}} = \left(\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}\right)^{-1}$

The lecture ends here

Questions of the Lecture 5

(L-5) Question 1. Use the Gauss-Jordan elimination method to invert

(a) $\mathbf{A}_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.

(b) $\mathbf{A}_2 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$.

(c) $\mathbf{A}_3 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$.

(Strang, 1988, exercise 6 from section 1.6.)

(L-5) Question 2.

(a) If $\mathbf{A}$ is invertible and $\mathbf{AB} = \mathbf{AC}$, prove quickly that $\mathbf{B} = \mathbf{C}$.
(b) If $\mathbf{A} = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$, find an example with $\mathbf{AB} = \mathbf{AC}$, but $\mathbf{B} \neq \mathbf{C}$.
(Strang, 1988, exercise 4 from section 1.6.)

(L-5) Question 3. Use the Gauss-Jordan method to invert the generic matrix 2×2

$$\mathbf{M} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

The matrix is invertible (not singular) only when ...

(L-5) Question 4. Use the Gauss-Jordan method to invert the following matrices.

$$\mathbf{A} = \begin{bmatrix} 3 & 1 & 2 \\ -1 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}; \qquad \mathbf{B} = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 4 & -2 \\ 1 & 3 & 1 \end{bmatrix}$$

(L-5) Question 5. If the 3 by 3 matrix $\mathbf{A}$ has $\mathbf{A}_{|1} + \mathbf{A}_{|2} = \mathbf{A}_{|3}$, show that $\mathbf{A}$ is not invertible by two different methods:

(a) Find a nonzero solution $\boldsymbol{x}$ to $\mathbf{A}\boldsymbol{x} = \mathbf{0}$.
(b) Elimination keeps $column\,1 + column\,2 = column\,3$. Explain why there is no third pivot.
(Strang, 1988, exercise 26 from section 1.6.)

(L-5) Question 6. Find the inverses of

(a) $\mathbf{A}_1 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & 3 & 0 & 0 \\ 4 & 0 & 0 & 0 \end{bmatrix}$.

(b) $\mathbf{A}_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1/2 & 1 & 0 & 0 \\ 0 & -2/3 & 1 & 0 \\ 0 & 0 & -3/4 & 1 \end{bmatrix}$.

(c) $\mathbf{A}_3 = \begin{bmatrix} a & b & 0 & 0 \\ c & d & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{bmatrix}$.

(Strang, 1988, exercise 10 from section 1.6.)

(L-5) Question 7. Find the inverse of

$$\mathbf{A} = \begin{bmatrix} a & b & b \\ a & a & b \\ a & a & a \end{bmatrix}$$

What values of a and b make the matrix singular?
(Strang, 1988, exercise 42 from section 1.6.)

(L-5) Question 8. Find $\mathbf{E}^2$, $\mathbf{E}^8$ and $\mathbf{E}^{-1}$ if $\mathbf{E} = \begin{bmatrix} 1 & 0 \\ 6 & 1 \end{bmatrix}$
(Strang, 1988, exercise 6 from section 1.5.)

(L-5) Question 9. Consider the following permutation matrix:

$$\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Find $\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}{}^{-1}$. Can you say something else about the relationship between $\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}$ and $\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}{}^{-1}$?

(L-5) Question 10. The 3 by 3 matrix $\mathbf{A}$ reduces to the identity matrix $\mathbf{I}$ by the following three column operations (in order):

$\underset{[(-4)\mathbf{1}+\mathbf{2}]}{\tau}$: Subtract 4 times column 1 from column 2.

$\underset{[(-3)\mathbf{1}+\mathbf{3}]}{\tau}$: Subtract 3 times column 1 from column 3.

$\underset{[(-1)\mathbf{3}+\mathbf{2}]}{\tau}$: Subtract column 3 from column 2.

(a) Write $\mathbf{A}^{-1}$ in terms of elementary matrices $\mathbf{I}_{\tau}$. **Then compute $\mathbf{A}^{-1}$.**
(b) What is the original matrix $\mathbf{A}$?
(Based on *MIT Course 18.06 Quiz 1, October 4, 2006*)

(L-5) Question 11. The 3 by 3 matrix $\mathbf{A}$ reduces to the identity matrix $\mathbf{I}$ by the following three **row** operations (in order):

$\underset{[(-4)\mathbf{1}+\mathbf{2}]}{\tau}$: Subtract 4 times row 1 from row 2.

$\underset{[(-3)\mathbf{1}+\mathbf{3}]}{\tau}$: Subtract 3 times row 1 from row 3.

$\underset{[(-1)\mathbf{3}+\mathbf{2}]}{\tau}$: Subtract row 3 from row 2.

(a) Write $\mathbf{A}^{-1}$ in terms of the $\mathbf{E}$'s. **Then compute $\mathbf{A}^{-1}$.**
(b) What is the original matrix $\mathbf{A}$?
(*MIT Course 18.06 Quiz 1, October 4, 2006*)

(L-5) Question 12.

(a) Find the inverse of $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$
(b) Find the inverse of the following matrix **using the Gauss-Jordan method**

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ a & b & c & d \end{bmatrix}$$

(Poole, 2004, exercise 36, 38 and 59 from section 3.3.)

(L-5) Question 13. Consider the squared matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$. True or false?

(a) If $\mathbf{AB} = \mathbf{I}$ and $\mathbf{CA} = \mathbf{I}$ then $\mathbf{B} = \mathbf{C}$.
(b) $(\mathbf{AB})^2 = \mathbf{A}^2\mathbf{B}^2$.

(L-5) QUESTION 14. Consider the matrix $\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 2 \\ 1 & a & 0 & 2a \\ a & 0 & 1 & 0 \\ 1 & 0 & a & 1 \end{bmatrix}$

(a) Prove that $\mathbf{A}$ is invertible for any value of a.
(b) Compute $\mathbf{A}^{-1}$ when $a = 0$.

(L-5) QUESTION 15. Consider the matrix $\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix}$. Find $\mathbf{A}^{-1}$.

(L-5) QUESTION 16. Find (if possible) the inverses of the following matrices

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}; \qquad \mathbf{B} = \begin{bmatrix} 1 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 1 \end{bmatrix}.$$

(L-5) QUESTION 17. There is a finite number ($n!$) of $n \times n$ permutation matrices. In addition, any power of a permutation matrix is also a permutation matrix. Use these facts to prove that $\left(\mathbf{I}_{\underset{[\mathfrak{S}]}{\boldsymbol{\tau}}}\right)^r = \mathbf{I}$ for some integer r.

End of Questions of the Lecture 5

References

Poole, D. (2004). *Álgebra lineal. Una introducción moderna.* Thomson Learning, Mexico D.F. ISBN 970-686-272-2.

Strang, G. (1988). *Linear algebra and its applications.* Thomson Learning, Inc., third ed. ISBN 0-15-551005-3.

Strang, G. (2003). *Introduction to Linear Algebra.* Wellesley-Cambridge Press, Wellesley, Massachusetts. USA, third ed. ISBN 0-9614088-9-8.

Solutions

(L-4) Question 1(a)

$$\mathbf{A} = \begin{bmatrix} 1 & 4 & -2 \\ 1 & 6 & 2 \\ 0 & 1 & 0 \end{bmatrix} \xrightarrow[\substack{[(-4)\mathbf{1}+\mathbf{2}] \\ [(2)\mathbf{1}+\mathbf{3}]}]{\boldsymbol{\tau}} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 4 \\ 0 & 1 & 0 \end{bmatrix} \xrightarrow[[(-2)\mathbf{2}+\mathbf{3}]]{\boldsymbol{\tau}} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 1 & -2 \end{bmatrix} = \mathbf{L}$$

therefore

$$\mathbf{I}_{\substack{\boldsymbol{\tau} \\ [(-4)\mathbf{1}+\mathbf{2}]}} = \begin{bmatrix} 1 & -4 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}; \quad \mathbf{I}_{\substack{\boldsymbol{\tau} \\ [(2)\mathbf{1}+\mathbf{3}]}} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}; \quad \mathbf{I}_{\substack{\boldsymbol{\tau} \\ [(-2)\mathbf{2}+\mathbf{3}]}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix};$$

```
A = Matrix([ [1, 4,-2], [1, 6, 2], [0, 1, 0] ])
L = Elim(A,1)
Tr = L.TrC
display(Tr)
Sistema([ Sistema([t, I(3)&t]) for t in Tr])
```

□

(L-4) Question 1(b)

$$\mathbf{E} = \mathbf{I}_{\substack{\boldsymbol{\tau} \\ [(-4)\mathbf{1}+\mathbf{2}] \\ [(2)\mathbf{1}+\mathbf{3}] \\ [(-2)\mathbf{2}+\mathbf{3}]}} = \left(\mathbf{I}_{\substack{\boldsymbol{\tau} \\ [(-4)\mathbf{1}+\mathbf{2}]}}\right)\left(\mathbf{I}_{\substack{\boldsymbol{\tau} \\ [(2)\mathbf{1}+\mathbf{3}]}}\right)\left(\mathbf{I}_{\substack{\boldsymbol{\tau} \\ [(-2)\mathbf{2}+\mathbf{3}]}}\right) = \begin{bmatrix} 1 & -4 & 10 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix};$$

$$\mathbf{AE} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 1 & -2 \end{bmatrix} = \mathbf{K}.$$

```
ME = Sistema( [ I(3) & t for t in Tr])
E  = (ME|1)*(ME|2)*(ME|3)
display(E)
display(A*E)
A & Tr
```

□

(L-4) Question 2. Subtracting 2 times the first column from the second and 4 times from the third, and then adding 2 times the second to the third

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 4 \\ -1 & -3 & -2 \\ 0 & 1 & c \end{bmatrix} \xrightarrow[\substack{[(-2)\mathbf{1}+\mathbf{2}] \\ [(-4)\mathbf{1}+\mathbf{3}]}]{\boldsymbol{\tau}} \begin{bmatrix} 1 & 0 & 0 \\ -1 & -1 & 2 \\ 0 & 1 & c \end{bmatrix} \xrightarrow[[(2)\mathbf{2}+\mathbf{3}]]{\boldsymbol{\tau}} \begin{bmatrix} 1 & 0 & 0 \\ -1 & -1 & 0 \\ 0 & 1 & c+2 \end{bmatrix} = \mathbf{L}$$

We will find only two pivots if the last column of $\mathbf{L}$ consists solely of zeros, which occurs when $c = -2$.

```
c = sympy.symbols('c')
Elim( Matrix([[1,2,4],[-1,-3,-2],[0,1,c]]) )
```

□

(L-4) Question 3(a) $\mathbf{E} = \left(\mathbf{I}_{\substack{\boldsymbol{\tau} \\ [(-1)\mathbf{1}+\mathbf{2}]}}\right)\left(\mathbf{I}_{\substack{\boldsymbol{\tau} \\ [\mathbf{2}\rightleftharpoons\mathbf{3}]}}\right) = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \mathbf{I}_{\substack{\boldsymbol{\tau} \\ [(-1)\mathbf{1}+\mathbf{2}] \\ [\mathbf{2}\rightleftharpoons\mathbf{3}]}}$

□

(L-4) Question 3(b) $\mathbf{N} = \left(\mathbf{I}_{\substack{\boldsymbol{\tau} \\ [\mathbf{2}\rightleftharpoons\mathbf{3}]}}\right)\left(\mathbf{I}_{\substack{\boldsymbol{\tau} \\ [(-1)\mathbf{1}+\mathbf{3}]}}\right) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \mathbf{I}_{\substack{\boldsymbol{\tau} \\ [\mathbf{2}\rightleftharpoons\mathbf{3}] \\ [(-1)\mathbf{1}+\mathbf{3}]}}$

The reason **E** and **N** are equal is that the operations they perform are equivalent: in the first case we subtract column 1 from column 2 and then put the result in the third column. In the second case, we move column 2 to the right and perform the subtraction there.

□

(L-4) Question 4.

$$\begin{bmatrix} 2 & 2 & 0 \\ 1 & 4 & 9 \\ 1 & 3 & 9 \end{bmatrix} \xrightarrow{[(-1)\overset{\boldsymbol{\tau}}{\mathbf{1}+\mathbf{2}}]} \begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 9 \\ 1 & 2 & 9 \end{bmatrix} \xrightarrow{[(-3)\overset{\boldsymbol{\tau}}{\mathbf{2}+\mathbf{3}}]} \begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 1 & 2 & 3 \end{bmatrix}$$

Therefore

$$\mathbf{I}_{\substack{\boldsymbol{\tau} \\ [(-1)\mathbf{1}+\mathbf{2}] \\ [(-3)\mathbf{2}+\mathbf{3}]}} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{E}.$$

```
A      = Matrix([[2,2,0],[1,4,9],[1,3,9]])
TrfCol = Elim(A).TrC
display(TrfCol)
I(3) & TrfCol
```

□

(L-4) Question 5. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow{[(1)\overset{\boldsymbol{\tau}}{\mathbf{1}+\mathbf{2}}]} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \xrightarrow{[(-1)\overset{\boldsymbol{\tau}}{\mathbf{2}+\mathbf{1}}]} \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix} \xrightarrow{[(1)\overset{\boldsymbol{\tau}}{\mathbf{1}+\mathbf{2}}]} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \xrightarrow{[(-1)\overset{\boldsymbol{\tau}}{\mathbf{1}}]} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

```
pasos = [ T((1,1,2)), T((-1,2,1)), T((1,1,2)),  T((-1,1))]
Math(rprElim(I(2),[[],pasos]))
# I(2) & T((1,1,2)) & T((-1,2,1)) & T((1,1,2)) & T((-1,1))
```

□

(L-4) Question 6(a)

$$\mathbf{I}_{\substack{\boldsymbol{\tau} \\ [(-5)\mathbf{1}+\mathbf{2}]}} = \begin{bmatrix} 1 & -5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

```
display( T((-5,1,2)) )
I(3) & T((-5,1,2))
```

□

(L-4) Question 6(b)

$$\mathbf{I}_{\substack{\boldsymbol{\tau} \\ [(-7)\mathbf{2}+\mathbf{3}]}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -7 \\ 0 & 0 & 1 \end{bmatrix}$$

```
display( T((-7,2,3)) )
I(3) & T((-7,2,3))
```

□

(L-4) Question 6(c)

$$\mathbf{I}_{\substack{\boldsymbol{\tau} \\ [\mathbf{1}\rightleftharpoons\mathbf{2}]}} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}; \qquad \mathbf{I}_{\substack{\boldsymbol{\tau} \\ [\mathbf{2}\rightleftharpoons\mathbf{3}]}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

The matrix that performs all the changes at once will be

$$\mathbf{I}_{\substack{\boldsymbol{\tau} \\ [\mathbf{1}\rightleftharpoons\mathbf{2}]\substack{\boldsymbol{\tau} \\ [\mathbf{2}\rightleftharpoons\mathbf{3}]}}} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \mathbf{I}_{\substack{\boldsymbol{\tau} \\ [\mathfrak{S}]}}$$

The product of matrices always yields a matrix whose columns are similar to those of the identity matrix, but generally reordered in a different arrangement; we call these matrices *permutation matrices* and denote them by: $\mathbf{I}_{\substack{\boldsymbol{\tau} \\ [\mathfrak{S}]}}$.

```
display( T([ {1,2}, {2,3} ]) )
I(3) & T([ {1,2}, {2,3} ])
```

□

(L-4) Question 7(a)

$$\mathbf{A}_{\underset{[(-5)\mathbf{1}+\mathbf{2}]}{\boldsymbol{\tau}}\underset{[(-7)\mathbf{2}+\mathbf{3}]}{\boldsymbol{\tau}}} = \begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}1 & -5 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\0 & 1 & -7\\0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}1 & -5 & 35\\0 & 1 & -7\\0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & -5 & 35\end{bmatrix}.$$

```
TrfC = T([ (-5,1,2), (-7,2,3) ])
display( I(3) & TrfC )
Matrix([ [1,0,0]  ]) & TrfC
```

□

(L-4) Question 7(b)

$$\mathbf{A}_{\underset{[(-7)\mathbf{1}+\mathbf{2}]}{\boldsymbol{\tau}}\underset{[(-5)\mathbf{2}+\mathbf{3}]}{\boldsymbol{\tau}}} = \begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\0 & 1 & -7\\0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & -5 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}1 & -5 & 0\\0 & 1 & -7\\0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & -5 & 0\end{bmatrix}.$$

```
TrfC = T([ (-7,2,3), (-5,1,2) ])
display( I(3) & TrfC )
Matrix([ [1,0,0]  ]) & TrfC
```

□

(L-4) Question 7(c) Then column 3 is not affected by column 1.

□

(L-4) Question 8. On one hand $\begin{pmatrix}1, & 0,\end{pmatrix}\mathbf{M} = \begin{pmatrix}0, & 1,\end{pmatrix} \quad \Rightarrow {}_{1|}\mathbf{M} = \begin{pmatrix}0, & 1,\end{pmatrix}$; on the other hand $\begin{pmatrix}0, & 1,\end{pmatrix}\mathbf{M} = \begin{pmatrix}1, & 0,\end{pmatrix} \quad \Rightarrow {}_{2|}\mathbf{M} = \begin{pmatrix}1, & 0,\end{pmatrix}$. Thus, the matrix is $\begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$.

□

(L-4) Question 9. If we multiply $\mathbf{A}\left(\mathbf{I}_{\underset{[i\rightleftharpoons j]}{\boldsymbol{\tau}}}\right)$, it exchanges the columns of $\mathbf{A}$, but if we multiply $\left({}_{\underset{[i\rightleftharpoons j]}{\boldsymbol{\tau}}}\mathbf{I}\right)\mathbf{A}$, it exchanges the rows. For example:

$$\left({}_{\underset{[\mathbf{1}\rightleftharpoons\mathbf{2}]}{\boldsymbol{\tau}}}\mathbf{I}\right)\mathbf{A} = \begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}\begin{bmatrix}a & b\\c & d\end{bmatrix} = \begin{bmatrix}c & d\\a & b\end{bmatrix}.$$

$$\mathbf{A}\left(\mathbf{I}_{\underset{[\mathbf{1}\rightleftharpoons\mathbf{2}]}{\boldsymbol{\tau}}}\right) = \begin{bmatrix}a & b\\c & d\end{bmatrix}\begin{bmatrix}0 & 1\\1 & 0\end{bmatrix} = \begin{bmatrix}b & a\\d & c\end{bmatrix}.$$

Note that $\left(\mathbf{I}_{\underset{[i\rightleftharpoons j]}{\boldsymbol{\tau}}}\right)$ is symmetric, so that $\left(\mathbf{I}_{\underset{[i\rightleftharpoons j]}{\boldsymbol{\tau}}}\right) = \left({}_{\underset{[i\rightleftharpoons j]}{\boldsymbol{\tau}}}\mathbf{I}\right)$.

```
a, b, c, d  = sympy.symbols('a b c d')
A = Matrix([ [a,b], [c,d] ])
display( (T({1,2}) & I(2)) * A )
display( A * (I(2) & T({1,2})) )
(I(2) & T({1,2})) == (T({1,2}) & I(2))
```

□

(L-4) Question 10. Assume that the first column is $\begin{pmatrix}a, & a, & a,\end{pmatrix}$, the second is $\begin{pmatrix}b, & b, & b,\end{pmatrix}$ and the third is

$\begin{pmatrix} c, & c, & c, \end{pmatrix}$; and where $a \neq 0$, thus

$$\begin{aligned}\mathbf{A}\boldsymbol{x} &= \begin{bmatrix} a & b & c \\ a & b & c \\ a & b & c \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_1 \begin{pmatrix} a \\ a \\ a \end{pmatrix} + x_2 \begin{pmatrix} b \\ b \\ b \end{pmatrix} + x_3 \begin{pmatrix} c \\ c \\ c \end{pmatrix} \\ &= ax_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + bx_2 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + cx_3 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = (ax_1 + bx_2 + cx_3) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\end{aligned}$$

If we apply Gaussian elimination, we obtain

$$\begin{bmatrix} a & b & c \\ a & b & c \\ a & b & c \end{bmatrix} \xrightarrow{\overset{\boldsymbol{\tau}}{\left[\left(-\frac{b}{a}\right)\mathbf{1}+\mathbf{2}\right]}} \begin{bmatrix} a & 0 & c \\ a & 0 & c \\ a & 0 & c \end{bmatrix} \xrightarrow{\overset{\boldsymbol{\tau}}{\left[\left(-\frac{c}{a}\right)\mathbf{1}+\mathbf{2}\right]}} \begin{bmatrix} a & 0 & 0 \\ a & 0 & 0 \\ a & 0 & 0 \end{bmatrix};$$

that is, only one pivot.

□

Exercise 1. Una permutación $\mathbf{I}_{\underset{[\mathfrak{S}]}{\boldsymbol{\tau}}}$ es el producto de k matrices intercambio A permutation $\mathbf{I}_{\underset{[\mathfrak{S}]}{\boldsymbol{\tau}}}$ is the product of k exchange matrices, $\mathbf{I}_{\underset{[\cdot\rightleftharpoons\cdot]}{\boldsymbol{\tau}}}$, that is, a sequence of k exchanges between the columns of the identity matrix, $\mathbf{I}_{\underset{[\mathfrak{S}]}{\boldsymbol{\tau}}} = \left(\mathbf{I}_{\underset{[h\rightleftharpoons i]}{\boldsymbol{\tau}}}\right)\left(\mathbf{I}_{\underset{[j\rightleftharpoons k]}{\boldsymbol{\tau}}}\right)\cdots\left(\mathbf{I}_{\underset{[p\rightleftharpoons q]}{\boldsymbol{\tau}}}\right)$, so its transpose is $\left(\mathbf{I}_{\underset{[\mathfrak{S}]}{\boldsymbol{\tau}}}\right)^{\mathsf{T}} = \left(_{\underset{[p\rightleftharpoons q]}{\boldsymbol{\tau}}}\mathbf{I}\right)\cdots\left(_{\underset{[j\rightleftharpoons k]}{\boldsymbol{\tau}}}\mathbf{I}\right)\left(_{\underset{[h\rightleftharpoons i]}{\boldsymbol{\tau}}}\mathbf{I}\right)$; where each exchange matrix is symmetric, $\left(_{\underset{[i\rightleftharpoons j]}{\boldsymbol{\tau}}}\mathbf{I}\right) = \left(\mathbf{I}_{\underset{[i\rightleftharpoons j]}{\boldsymbol{\tau}}}\right)$ and its square is the identity matrix (if the same columns are exchanged twice, one returns to the starting point), $\left(\mathbf{I}_{\underset{[i\rightleftharpoons j]}{\boldsymbol{\tau}}}\right)\left(\mathbf{I}_{\underset{[i\rightleftharpoons j]}{\boldsymbol{\tau}}}\right) = \mathbf{I}$. Thus

$$\left(\mathbf{I}_{\underset{[\mathfrak{S}]}{\boldsymbol{\tau}}}\right)^{\mathsf{T}}\left(\mathbf{I}_{\underset{[\mathfrak{S}]}{\boldsymbol{\tau}}}\right) = \left(_{\underset{[p\rightleftharpoons q]}{\boldsymbol{\tau}}}\mathbf{I}\right)\cdots\overbrace{\left(_{\underset{[j\rightleftharpoons k]}{\boldsymbol{\tau}}}\mathbf{I}\right)\underbrace{\left(_{\underset{[h\rightleftharpoons i]}{\boldsymbol{\tau}}}\mathbf{I}\right)\left(\mathbf{I}_{\underset{[h\rightleftharpoons i]}{\boldsymbol{\tau}}}\right)}_{\mathbf{I}}\left(\mathbf{I}_{\underset{[j\rightleftharpoons k]}{\boldsymbol{\tau}}}\right)}^{\mathbf{I}}\cdots\left(\mathbf{I}_{\underset{[p\rightleftharpoons q]}{\boldsymbol{\tau}}}\right) = \mathbf{I}.$$

□

(L-5) Question 1(a) $\left[\begin{array}{c}\mathbf{A}_1 \\ \hline \mathbf{I}\end{array}\right] = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \\ \hline 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow{\overset{\boldsymbol{\tau}}{\substack{[(-1)\mathbf{2}+\mathbf{1}] \\ [(-1)\mathbf{2}+\mathbf{3}]}}} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \hline 1 & 0 & 0 \\ -1 & 1 & -1 \\ 0 & 0 & 1 \end{array}\right].$

Therefore $\mathbf{A}_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

```
A1 = Matrix([[1,0,0],[1,1,1],[0,0,1]])
InvMat(A1, 1)
```

□

(L-5) Question 1(b)

$$\left[\begin{array}{c}\mathbf{A}_2\\ \hline \mathbf{I}\end{array}\right] = \left[\begin{array}{ccc} 2 & -1 & 0\\ -1 & 2 & -1\\ 0 & -1 & 2\\ \hline 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{array}\right] \xrightarrow[\left[(1)\mathbf{2}+\mathbf{1}\right]]{\boldsymbol{\tau}\ \left[(2)\mathbf{2}\right]} \left[\begin{array}{ccc} 2 & 0 & 0\\ -1 & 3 & -1\\ 0 & -2 & 2\\ \hline 1 & 1 & 0\\ 0 & 2 & 0\\ 0 & 0 & 1\end{array}\right] \xrightarrow[\left[(1)\mathbf{2}+\mathbf{3}\right]]{\boldsymbol{\tau}\ \left[(3)\mathbf{3}\right]} \left[\begin{array}{ccc} 2 & 0 & 0\\ -1 & 3 & 0\\ 0 & -2 & 4\\ \hline 1 & 1 & 1\\ 0 & 2 & 2\\ 0 & 0 & 3\end{array}\right] \xrightarrow[\left[(1)\mathbf{3}+\mathbf{2}\right]]{\boldsymbol{\tau}\ \left[(2)\mathbf{2}\right]}$$

$$\left[\begin{array}{ccc} 2 & 0 & 0\\ -1 & 6 & 0\\ 0 & 0 & 4\\ \hline 1 & 3 & 1\\ 0 & 6 & 2\\ 0 & 3 & 3\end{array}\right] \xrightarrow[\left[(1)\mathbf{2}+\mathbf{1}\right]]{\boldsymbol{\tau}\ \left[(6)\mathbf{1}\right]} \left[\begin{array}{ccc} 12 & 0 & 0\\ 0 & 6 & 0\\ 0 & 0 & 4\\ \hline 9 & 3 & 1\\ 6 & 6 & 2\\ 3 & 3 & 3\end{array}\right] \xrightarrow[\left[\left(\frac{2}{3}\right)\mathbf{2}\right]]{\boldsymbol{\tau}\ \left[\left(\frac{1}{3}\right)\mathbf{1}\right]} \left[\begin{array}{ccc} 4 & 0 & 0\\ 0 & 4 & 0\\ 0 & 0 & 4\\ \hline 3 & 2 & 1\\ 2 & 4 & 2\\ 1 & 2 & 3\end{array}\right] \xrightarrow[\left[\left(\frac{1}{4}\right)\mathbf{3}\right]]{\boldsymbol{\tau}\ \left[\left(\frac{1}{4}\right)\mathbf{1}\right]\ \left[\left(\frac{1}{4}\right)\mathbf{2}\right]} \frac{1}{4}\left[\begin{array}{c} 4\mathbf{I}\\ \hline \begin{array}{ccc} 3 & 2 & 1\\ 2 & 4 & 2\\ 1 & 2 & 3\end{array}\end{array}\right]$$

Therefore $\mathbf{A}_2^{-1} = \frac{1}{4}\begin{bmatrix} 3 & 2 & 1\\ 2 & 4 & 2\\ 1 & 2 & 3\end{bmatrix}$

□

(L-5) Question 1(c) $\left[\begin{array}{c}\mathbf{A}_3\\ \hline \mathbf{I}\end{array}\right] = \left[\begin{array}{ccc} 0 & 0 & 1\\ 0 & 1 & 1\\ 1 & 1 & 1\\ \hline 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{array}\right] \xrightarrow[\left[\mathbf{1}\rightleftarrows\mathbf{3}\right]]{\boldsymbol{\tau}} \left[\begin{array}{ccc} 1 & 0 & 0\\ 1 & 1 & 0\\ 1 & 1 & 1\\ \hline 0 & 0 & 1\\ 0 & 1 & 0\\ 1 & 0 & 0\end{array}\right] \xrightarrow[\left[(-1)\mathbf{3}+\mathbf{2}\right]]{\boldsymbol{\tau}\ \left[(-1)\mathbf{2}+\mathbf{1}\right]} \left[\begin{array}{ccc} 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\\ \hline 0 & -1 & 1\\ -1 & 1 & 0\\ 1 & 0 & 0\end{array}\right].$

Therefore $\mathbf{A}_3^{-1} = \begin{bmatrix} 0 & -1 & 1\\ -1 & 1 & 0\\ 1 & 0 & 0\end{bmatrix}$

□

(L-5) Question 2(a)

$$\begin{aligned} \mathbf{AB} &= \mathbf{AC}\\ \mathbf{A}^{-1}\mathbf{AB} &= \mathbf{A}^{-1}\mathbf{AC}\\ \mathbf{IB} &= \mathbf{IC}\\ \mathbf{B} &= \mathbf{C}. \end{aligned}$$

□

(L-5) Question 2(b)

$$\begin{bmatrix} 1 & 0\\ 0 & 0\end{bmatrix}\begin{bmatrix} 0 & 0\\ 5 & 5\end{bmatrix} = \begin{bmatrix} 0 & 0\\ 0 & 0\end{bmatrix} = \begin{bmatrix} 1 & 0\\ 0 & 0\end{bmatrix}\begin{bmatrix} 0 & 0\\ 2 & 2\end{bmatrix}$$

□

(L-5) Question 3.

$$\left[\begin{array}{cc} a & b\\ c & d\\ \hline 1 & 0\\ 0 & 1\end{array}\right] \xrightarrow[\left[\left(\frac{-b}{a}\right)\mathbf{1}+\mathbf{2}\right]]{\boldsymbol{\tau}} \left[\begin{array}{cc} a & 0\\ c & d-\frac{bc}{a}\\ \hline 1 & -b/a\\ 0 & 1\end{array}\right] = \left[\begin{array}{cc} a & 0\\ c & \frac{ad-bc}{a}\\ \hline 1 & -b/a\\ 0 & 1\end{array}\right] \xrightarrow[\left[\left(\frac{a}{ad-bc}\right)\mathbf{2}\right]]{\boldsymbol{\tau}} \left[\begin{array}{cc} a & 0\\ c & 1\\ \hline 1 & \frac{-b}{ad-bc}\\ 0 & \frac{a}{ad-bc}\end{array}\right]$$

$$\xrightarrow[\left[(-c)\mathbf{2}+\mathbf{1}\right]]{\boldsymbol{\tau}} \left[\begin{array}{cc} a & 0\\ 0 & 1\\ \hline 1+\frac{bc}{ad-bc} & \frac{-b}{ad-bc}\\ \frac{-ac}{ad-bc} & \frac{a}{ad-bc}\end{array}\right] = \left[\begin{array}{cc} a & 0\\ 0 & 1\\ \hline \frac{ad}{ad-bc} & \frac{-b}{ad-bc}\\ \frac{-ac}{ad-bc} & \frac{a}{ad-bc}\end{array}\right] \xrightarrow[\left[\left(\frac{1}{a}\right)\mathbf{2}\right]]{\boldsymbol{\tau}} \left[\begin{array}{cc} 1 & 0\\ 0 & 1\\ \hline \frac{d}{ad-bc} & \frac{-b}{ad-bc}\\ \frac{-c}{ad-bc} & \frac{a}{ad-bc}\end{array}\right]$$

so

$$\mathbf{M}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b\\ -c & a\end{bmatrix}$$

The matrix is invertible only when $ad \neq bc$.

```
a, b, c, d = sympy.symbols('a b c d')
A = Matrix([ [a,b], [c,d] ])
InvMat(A, 1)
```

□

(L-5) Question 4.

$$\left[\begin{array}{ccc} 3 & 1 & 2 \\ -1 & 0 & 1 \\ 0 & 1 & 6 \\ \hline 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[{[1 \rightleftharpoons 2]}]{\tau} \left[\begin{array}{ccc} 1 & 3 & 2 \\ 0 & -1 & 1 \\ 1 & 0 & 6 \\ \hline 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{[(-3)\mathbf{1}+\mathbf{2}] \\ [(-2)\mathbf{1}+\mathbf{3}]}]{\tau} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 1 & -3 & 4 \\ \hline 0 & 1 & 0 \\ 1 & -3 & -2 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{[(-1)\mathbf{2}] \\ [(-1)\mathbf{2}+\mathbf{3}]}]{\tau} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 3 & 1 \\ \hline 0 & -1 & 1 \\ 1 & 3 & -5 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{[(-3)\mathbf{3}+\mathbf{2}] \\ [(-1)\mathbf{3}+\mathbf{1}]}]{\tau} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \hline -1 & -4 & 1 \\ 6 & 18 & -5 \\ -1 & -3 & 1 \end{array}\right] = \left[\begin{array}{c} \mathbf{I} \\ \hline \mathbf{A}^{-1} \end{array}\right]$$

$$\left[\begin{array}{c} \mathbf{B} \\ \hline \mathbf{I} \end{array}\right] = \left[\begin{array}{ccc} 1 & 2 & 1 \\ -1 & 4 & -2 \\ 1 & 3 & 1 \\ \hline 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{[(-2)\mathbf{1}+\mathbf{2}] \\ [(-1)\mathbf{1}+\mathbf{3}]}]{\tau} \left[\begin{array}{ccc} 1 & 0 & 0 \\ -1 & 6 & -1 \\ 1 & 1 & 0 \\ \hline 1 & -2 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{[\mathbf{2} \rightleftharpoons \mathbf{3}] \\ [(-1)\mathbf{2}]}]{\tau} \left[\begin{array}{ccc} 1 & 0 & 0 \\ -1 & 1 & 6 \\ 1 & 0 & 1 \\ \hline 1 & 1 & -2 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{array}\right]$$

$$\xrightarrow[\substack{[(1)\mathbf{2}+\mathbf{1}] \\ [(-6)\mathbf{2}+\mathbf{3}]}]{\tau} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \\ \hline 2 & 1 & -8 \\ 0 & 0 & 1 \\ -1 & -1 & 6 \end{array}\right] \xrightarrow[{[(-1)\mathbf{3}+\mathbf{1}]}]{\tau} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \hline 10 & 1 & -8 \\ -1 & 0 & 1 \\ -7 & -1 & 6 \end{array}\right] = \left[\begin{array}{c} \mathbf{I} \\ \hline \mathbf{B}^{-1} \end{array}\right]$$

□

(L-5) Question 5(a)

$$\begin{bmatrix} a_1 & b_1 & a_1+b_1 \\ a_2 & b_2 & a_2+b_2 \\ a_3 & b_3 & a_3+b_3 \end{bmatrix} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

□

(L-5) Question 5(b) When we subtract the first and second columns from the third, we get a zero column (no pivot in that column)

$$\begin{bmatrix} a_1 & b_1 & a_1+b_1 \\ a_2 & b_2 & a_2+b_2 \\ a_3 & b_3 & a_3+b_3 \end{bmatrix} \xrightarrow[\substack{[(-1)\mathbf{1}+\mathbf{3}] \\ [(-1)\mathbf{2}+\mathbf{3}]}]{\tau} \begin{bmatrix} a_1 & b_1 & 0 \\ a_2 & b_2 & 0 \\ a_3 & b_3 & 0 \end{bmatrix}$$

□

(L-5) Question 6(a)

$$\left[\begin{array}{c} \mathbf{A}_1 \\ \hline \mathbf{I} \end{array}\right] = \left[\begin{array}{cccc} 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & 3 & 0 & 0 \\ 4 & 0 & 0 & 0 \\ \hline 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{[\mathbf{1} \rightleftharpoons \mathbf{4}] \\ [\mathbf{2} \rightleftharpoons \mathbf{3}]}]{\tau} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \\ \hline 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{array}\right] \xrightarrow[\substack{[(1/2)\mathbf{2}] \\ [(1/3)\mathbf{3}] \\ [(1/4)\mathbf{4}]}]{\tau} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \hline 0 & 0 & 0 & 1/4 \\ 0 & 0 & 1/3 & 0 \\ 0 & 1/2 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{array}\right]$$

□

(L-5) Question 6(b)

$$\begin{bmatrix}\mathbf{A}_2\\ \hline \mathbf{I}\end{bmatrix} = \left[\begin{array}{rrrr} 1 & 0 & 0 & 0\\ -1/2 & 1 & 0 & 0\\ 0 & -2/3 & 1 & 0\\ 0 & 0 & -3/4 & 1\\ \hline 1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{array}\right] \xrightarrow[\substack{[(3/4)\mathbf{4}+\mathbf{3}]\\ [(2/3)\mathbf{3}+\mathbf{2}]\\ [(1/2)\mathbf{2}+\mathbf{1}]}]{\boldsymbol{\tau}} \left[\begin{array}{rrrr} 1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\\ \hline 1 & 0 & 0 & 0\\ 1/2 & 1 & 0 & 0\\ 1/3 & 2/3 & 1 & 0\\ 1/4 & 1/2 & 3/4 & 1\end{array}\right] = \begin{bmatrix}\mathbf{I}\\ \hline \mathbf{A}_2^{-1}\end{bmatrix}$$

□

(L-5) Question 6(c) We just need to repeat, for each block, the steps of Exercise 3:

$$\left[\begin{array}{cccc} a & b & 0 & 0\\ c & d & 0 & 0\\ 0 & 0 & a & b\\ 0 & 0 & c & d\\ \hline 1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{array}\right] \xrightarrow[\substack{[(-b/a)\mathbf{1}+\mathbf{2}]\\ [(\frac{a}{ad-bc})\mathbf{2}]\\ [(-c)\mathbf{2}+\mathbf{1}]\\ [(\frac{1}{a})\mathbf{2}]}]{\boldsymbol{\tau}} \left[\begin{array}{cccc} 1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & a & b\\ 0 & 0 & c & d\\ \hline \frac{d}{ad-bc} & \frac{-b}{ad-bc} & 0 & 0\\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{array}\right] \xrightarrow[\substack{[(-b/a)\mathbf{3}+\mathbf{4}]\\ [(\frac{a}{ad-bc})\mathbf{4}]\\ [(-c)\mathbf{4}+\mathbf{3}]\\ [(\frac{1}{a})\mathbf{4}]}]{\boldsymbol{\tau}} \left[\begin{array}{cccc} 1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\\ \hline \frac{d}{ad-bc} & \frac{-b}{ad-bc} & 0 & 0\\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} & 0 & 0\\ 0 & 0 & \frac{d}{ad-bc} & \frac{-b}{ad-bc}\\ 0 & 0 & \frac{-c}{ad-bc} & \frac{a}{ad-bc}\end{array}\right] = \begin{bmatrix}\mathbf{I}\\ \hline \mathbf{A}_3^{-1}\end{bmatrix}$$

□

(L-5) Question 7.

$$\left[\begin{array}{ccc} a & b & b\\ a & a & b\\ a & a & a\\ \hline 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{array}\right] \xrightarrow[\substack{[(\frac{-b}{a})\mathbf{1}+\mathbf{2}]\\ [(\frac{-b}{a})\mathbf{1}+\mathbf{3}]}]{\boldsymbol{\tau}} \left[\begin{array}{ccc} a & 0 & 0\\ a & a-b & 0\\ a & a-b & a-b\\ \hline 1 & \frac{-b}{a} & \frac{-b}{a}\\ 0 & 1 & 0\\ 0 & 0 & 1\end{array}\right] \xrightarrow[\substack{[(\frac{1}{a})\mathbf{1}]\\ [(\frac{1}{a-b})\mathbf{2}]\\ [(\frac{1}{a-b})\mathbf{3}]}]{\boldsymbol{\tau}} \left[\begin{array}{ccc} 1 & 0 & 0\\ 1 & 1 & 0\\ 1 & 1 & 1\\ \hline \frac{1}{a} & \frac{-b/a}{a-b} & \frac{-b/a}{a-b}\\ 0 & \frac{1}{a-b} & 0\\ 0 & 0 & \frac{1}{a-b}\end{array}\right] \xrightarrow[\substack{[(-1)\mathbf{2}+\mathbf{1}]\\ [(-1)\mathbf{3}+\mathbf{2}]}]{\boldsymbol{\tau}} \left[\begin{array}{ccc} 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\\ \hline \frac{1}{a-b} & 0 & \frac{-b/a}{a-b}\\ \frac{-1}{a-b} & \frac{1}{a-b} & 0\\ 0 & \frac{-1}{a-b} & \frac{1}{a-b}\end{array}\right]$$

Hence $\mathbf{A}^{-1} = \frac{1}{a-b}\begin{bmatrix} 1 & 0 & -b/a\\ -1 & 1 & 0\\ 0 & -1 & 1\end{bmatrix}$

There is no inverse when $a = b$, since the denominator $(a-b)$ is zero (in that case all rows in $\mathbf{A}$ are the same; and therefore there is only one pivot).

There is not inverse, either, when $a = 0$; since the third element on the first row is not defined (in that case $\mathbf{A}$ has a zero column and a zero row).

□

(L-5) Question 8. Since $\mathbf{E}$ is an elementary matrix

$$\mathbf{E}^2 = \begin{bmatrix} 1 & 0\\ 12 & 1\end{bmatrix}; \qquad \mathbf{E}^8 = \begin{bmatrix} 1 & 0\\ 48 & 1\end{bmatrix}; \qquad \mathbf{E}^{-1} = \begin{bmatrix} 1 & 0\\ -6 & 1\end{bmatrix}; \qquad \mathbf{E}^n = \begin{bmatrix} 1 & 0\\ 6n & 1\end{bmatrix}$$

□

(L-5) Question 9. $\begin{bmatrix} 0 & 0 & 0 & 1\\ 0 & 1 & 0 & 0\\ 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0\end{bmatrix}$. It's also its transpose: $\left(\mathbf{I}\underset{[\mathfrak{S}]}{\boldsymbol{\tau}}\right)^{\mathsf{T}}$.

□

(L-5) Question 10(a) Since $\mathbf{A}\left(\mathbf{I}_{\substack{\tau\\[(-4)\mathbf{1}+\mathbf{2}]}\substack{\tau\\[(-3)\mathbf{1}+\mathbf{3}]}\substack{\tau\\[(-1)\mathbf{3}+\mathbf{2}]}}\right) = \mathbf{I}$, then $\left(\mathbf{I}_{\substack{\tau\\[(-4)\mathbf{1}+\mathbf{2}]}\substack{\tau\\[(-3)\mathbf{1}+\mathbf{3}]}\substack{\tau\\[(-1)\mathbf{3}+\mathbf{2}]}}\right) = \mathbf{A}^{-1}$:

$$\mathbf{I}_{\substack{\tau\\[(-4)\mathbf{1}+\mathbf{2}]\\[(-3)\mathbf{1}+\mathbf{3}]\\[(-1)\mathbf{3}+\mathbf{2}]}} = \begin{bmatrix} 1 & -4 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & -3 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \mathbf{A}^{-1}$$

```
Tr = T((-4,1,2)) & T((-3,1,3)) & T((-1,3,2))
display(Tr)
Ainv = I(3) & Tr
Ainv
```

□

(L-5) Question 10(b) Since $\left(\mathbf{I}_{\substack{\tau\\[(-4)\mathbf{1}+\mathbf{2}]}\substack{\tau\\[(-3)\mathbf{1}+\mathbf{3}]}\substack{\tau\\[(-1)\mathbf{3}+\mathbf{2}]}}\right)^{-1} = \mathbf{A}$ then $\mathbf{A} = \left(\mathbf{I}_{\substack{\tau\\[(1)\mathbf{3}+\mathbf{2}]}\substack{\tau\\[(3)\mathbf{1}+\mathbf{3}]}\substack{\tau\\[(4)\mathbf{1}+\mathbf{2}]}}\right)$.

$$\mathbf{I}_{\substack{\tau\\[(1)\mathbf{3}+\mathbf{2}]\\[(3)\mathbf{1}+\mathbf{3}]\\[(4)\mathbf{1}+\mathbf{2}]}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \mathbf{A}$$

$$\text{Check:} \quad \mathbf{A}^{-1}\mathbf{A} = \begin{bmatrix} 1 & -1 & -3 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

```
display(Tr**-1)
A = I(3) & Tr**-1
Sistema([A, A*Ainv] )
```

□

(L-5) Question 11(a) Remember that ${}_{\tau}\mathbf{I}$ is the transpose of $\mathbf{I}_{\tau}$.

Since $\left({}_{\substack{\tau\\[(-1)\mathbf{3}+\mathbf{2}]}}\substack{\tau\\[(-3)\mathbf{1}+\mathbf{3}]}\substack{\tau\\[(-4)\mathbf{1}+\mathbf{2}]}\mathbf{I}\right)\mathbf{A} = \mathbf{I}$, then $\left({}_{\substack{\tau\\[(-1)\mathbf{3}+\mathbf{2}]}}\substack{\tau\\[(-3)\mathbf{1}+\mathbf{3}]}\substack{\tau\\[(-4)\mathbf{1}+\mathbf{2}]}\mathbf{I}\right) = \mathbf{A}^{-1}$:

$${}_{\substack{\tau\\[(-1)\mathbf{3}+\mathbf{2}]\\[(-3)\mathbf{1}+\mathbf{3}]\\[(-4)\mathbf{1}+\mathbf{2}]}}\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & -1 \\ -3 & 0 & 1 \end{bmatrix} = \mathbf{A}^{-1}$$

```
Tr   = T((-4,1,2)) & T((-3,1,3)) & T((-1,3,2))
display(Tr)
display( Sistema([t & I(3) for t in ~Tr]) )
Ainv = ~Tr & I(3)
Ainv
```

□

(L-5) Question 11(b) Since $\left({}_{\substack{\tau\\[(-1)\mathbf{3}+\mathbf{2}]}}\substack{\tau\\[(-3)\mathbf{1}+\mathbf{3}]}\substack{\tau\\[(-4)\mathbf{1}+\mathbf{2}]}\mathbf{I}\right)^{-1} = \mathbf{A}$ then $\mathbf{A} = \left({}_{\substack{\tau\\[(4)\mathbf{1}+\mathbf{2}]}}\substack{\tau\\[(3)\mathbf{1}+\mathbf{3}]}\substack{\tau\\[(1)\mathbf{3}+\mathbf{2}]}\mathbf{I}\right)$.

$${}_{\substack{\tau\\[(4)\mathbf{1}+\mathbf{2}]\\[(3)\mathbf{1}+\mathbf{3}]\\[(1)\mathbf{3}+\mathbf{2}]}}\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 1 \\ 3 & 0 & 1 \end{bmatrix} = \mathbf{A}$$

$$\text{Check:} \quad \mathbf{A}^{-1}\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & -1 \\ -3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 1 \\ 3 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

```
display(~Tr**-1)
display( Sistema([t & I(3) for t in ~Tr**-1]) )
A    = ~Tr**-1 & I(3)
Sistema([A, Ainv*A] )
```

□

(L-5) Question 12(a) The first one is an elementary matrix, its inverse is $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{bmatrix}$; the second one is a permutation matrix, its inverse is its transpose.

□

(L-5) Question 12(b)

$$\left[\begin{array}{c}\mathbf{A}\\ \hline \mathbf{I}\end{array}\right] = \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ a & b & c & d \\ \hline 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{\overset{\boldsymbol{\tau}}{[(\frac{1}{d})\mathbf{4}]}} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ a & b & c & 1 \\ \hline 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \frac{1}{d} \end{array}\right] \xrightarrow{\overset{\boldsymbol{\tau}}{\substack{[(-c)\mathbf{4}+\mathbf{3}]\\ [(-b)\mathbf{4}+\mathbf{2}]\\ [(-a)\mathbf{4}+\mathbf{1}]}}} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \hline 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\frac{a}{d} & -\frac{b}{d} & -\frac{c}{d} & \frac{1}{d} \end{array}\right] = \left[\begin{array}{c}\mathbf{I}\\ \hline \mathbf{A}^{-1}\end{array}\right]$$

```
a, b, c, d = sympy.symbols('a b c d')
C = Matrix([ [1,0,0,0], [0,1,0,0], [0,0,1,0], [a,b,c,d] ])
C.apila(I(4),1) & T((fracc(1,d),4)) & T([(-c,4,3),(-b,4,2),(-a,4,1)])
```

□

(L-5) Question 13(a) True, since

$$\mathbf{AB} = \mathbf{I} \quad \Rightarrow \quad \mathbf{B} = \mathbf{A}^{-1}.$$

and

$$\mathbf{CA} = \mathbf{I} \quad \Rightarrow \quad \mathbf{C} = \mathbf{A}^{-1}.$$

Therefore $\mathbf{B}$ and $\mathbf{C}$ are the same matrix $\mathbf{A}^{-1}$.

□

(L-5) Question 13(b) False:

$$(\mathbf{AB})^2 = (\mathbf{AB})\,(\mathbf{AB}) = \mathbf{ABAB}$$

is in general different from

$$\mathbf{A}^2\mathbf{B}^2 = \mathbf{AABB}.$$

Example:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}; \quad \mathbf{B} = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}. \qquad \mathbf{ABAB} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}; \quad \mathbf{AABB} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

□

(L-5) Question 14(a) Let us check if the rank is 4 (using column reduction):

$$\left[\begin{array}{c}\mathbf{A}\\ \hline \mathbf{I}\end{array}\right] = \left[\begin{array}{cccc} 0 & 1 & 0 & 2 \\ 1 & a & 0 & 2a \\ a & 0 & 1 & 0 \\ 1 & 0 & a & 1 \\ \hline 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{\overset{\boldsymbol{\tau}}{[(-2)\mathbf{2}+\mathbf{4}]}} \left[\begin{array}{cccc} 0 & 1 & 0 & 0 \\ 1 & a & 0 & 0 \\ a & 0 & 1 & 0 \\ 1 & 0 & a & 1 \\ \hline 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{\overset{\boldsymbol{\tau}}{[\mathbf{1}\rightleftharpoons\mathbf{2}]}} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ a & 1 & 0 & 0 \\ 0 & a & 1 & 0 \\ 0 & 1 & a & 1 \\ \hline 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] = \left[\begin{array}{c}\mathbf{L}\\ \hline \mathbf{E}\end{array}\right]$$

This matrix has rank 4 for any value of a; therefore it is invertible.

□

(L-5) Question 14(b)

$$\left[\begin{array}{c}\mathbf{A}\\ \hline \mathbf{I}\end{array}\right] \to \left[\begin{array}{c}\mathbf{L}\\ \hline \mathbf{E}\end{array}\right] = \left[\begin{array}{cccc} 1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 1 & 0 & 1\\ \hline 0 & 1 & 0 & 0\\ 1 & 0 & 0 & -2\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{array}\right] \xrightarrow[\left[(-1)\mathbf{4}+\mathbf{2}\right]]{\boldsymbol{\tau}} \left[\begin{array}{cccc} 1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\\ \hline 0 & 1 & 0 & 0\\ 1 & 2 & 0 & -2\\ 0 & 0 & 1 & 0\\ 0 & -1 & 0 & 1\end{array}\right] = \left[\begin{array}{c}\mathbf{I}\\ \hline \mathbf{A}^{-1}\end{array}\right]$$

```
a    = sympy.symbols('a')
A    = Matrix([ [0,1,0,2],[1,a,0,2*a],[a,0,1,0],[1,0,a,1] ])
Ainv = InvMat(A, 1)
Ainv.subs([(a,0)])
```

□

(L-5) Question 15. $\mathbf{A}^{-1} = \begin{bmatrix} 1 & -1 & 0 & 1\\ 0 & 1 & 0 & -\frac{1}{2}\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & -\frac{1}{2}\end{bmatrix}$

```
A = Matrix([ [1,1,0,1], [0,1,0,-1], [0,0,1,0], [0,0,0,-2] ])
InvMat(A, 1)
```

□

(L-5) Question 16.

$$\left[\begin{array}{c}\mathbf{A}\\ \hline \mathbf{I}\end{array}\right] = \left[\begin{array}{ccc} 1 & 0 & 1\\ 1 & 1 & 0\\ 0 & 1 & 1\\ \hline 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{array}\right] \xrightarrow[\left[(-1)\mathbf{1}+\mathbf{3}\right]]{\boldsymbol{\tau}} \left[\begin{array}{ccc} 1 & 0 & 0\\ 1 & 1 & -1\\ 0 & 1 & 1\\ \hline 1 & 0 & -1\\ 0 & 1 & 0\\ 0 & 0 & 1\end{array}\right] \xrightarrow[\substack{\left[(-1)\mathbf{2}+\mathbf{1}\right]\\ \left[(1)\mathbf{2}+\mathbf{3}\right]}]{\boldsymbol{\tau}} \left[\begin{array}{ccc} 1 & 0 & 0\\ 0 & 1 & 0\\ -1 & 1 & 2\\ \hline 1 & 0 & -1\\ -1 & 1 & 1\\ 0 & 0 & 1\end{array}\right] \xrightarrow[\substack{\left[\left(\frac{1}{2}\right)\mathbf{3}\right]\\ \left[(-1)\mathbf{3}+\mathbf{2}\right]\\ \left[(1)\mathbf{3}+\mathbf{1}\right]}]{\boldsymbol{\tau}} \left[\begin{array}{ccc} 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\\ \hline 1/2 & 1/2 & -1/2\\ -1/2 & 1/2 & 1/2\\ 1/2 & -1/2 & 1/2\end{array}\right] = \left[\begin{array}{c}\mathbf{I}\\ \hline \mathbf{A}^{-1}\end{array}\right]$$

$$\left[\begin{array}{c}\mathbf{B}\\ \hline \mathbf{I}\end{array}\right] = \left[\begin{array}{ccc} 1 & 1 & -2\\ 1 & -2 & 1\\ -2 & 1 & 1\\ \hline 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{array}\right] \xrightarrow[\substack{\left[(-1)\mathbf{1}+\mathbf{2}\right]\\ \left[(2)\mathbf{1}+\mathbf{3}\right]}]{\boldsymbol{\tau}} \left[\begin{array}{ccc} 1 & 0 & 0\\ 1 & -3 & 3\\ -2 & 3 & -3\\ \hline 1 & -1 & 2\\ 0 & 1 & 0\\ 0 & 0 & 1\end{array}\right] \xrightarrow[\left[(1)\mathbf{2}+\mathbf{3}\right]]{\boldsymbol{\tau}} \left[\begin{array}{ccc} 1 & 0 & 0\\ 1 & -3 & 0\\ -2 & 3 & 0\\ \hline 1 & -1 & 1\\ 0 & 1 & 1\\ 0 & 0 & 1\end{array}\right] = \left[\begin{array}{c}\mathbf{L}\\ \hline \mathbf{E}\end{array}\right]$$

The matrix $\mathbf{B}$ is singular (it has no inverse).

□

(L-5) Question 17. Since any power of a permutation matrix is a permutation matrix, and there is a finite number of permutation matrices; the sequence $\left(\mathbf{I}_{\substack{\boldsymbol{\tau}\\ [\mathfrak{S}]}}\right)$, $\left(\mathbf{I}_{\substack{\boldsymbol{\tau}\\ [\mathfrak{S}]}}\right)^2$, $\left(\mathbf{I}_{\substack{\boldsymbol{\tau}\\ [\mathfrak{S}]}}\right)^3$, $\ldots$, $\left(\mathbf{I}_{\substack{\boldsymbol{\tau}\\ [\mathfrak{S}]}}\right)^r$, $\ldots$ must repeat itself. Therefore, for some m and n, it must hold that $\left(\mathbf{I}_{\substack{\boldsymbol{\tau}\\ [\mathfrak{S}]}}\right)^m = \left(\mathbf{I}_{\substack{\boldsymbol{\tau}\\ [\mathfrak{S}]}}\right)^n$. And since permutation matrices are invertible, there exists $\left(\mathbf{I}_{\substack{\boldsymbol{\tau}\\ [\mathfrak{S}]}}\right)^{-n}$. Pre-multiplying by this matrix, we obtain:

$$\begin{aligned}\left(\mathbf{I}_{\substack{\boldsymbol{\tau}\\ [\mathfrak{S}]}}\right)^m &= \left(\mathbf{I}_{\substack{\boldsymbol{\tau}\\ [\mathfrak{S}]}}\right)^n\\ \left(\mathbf{I}_{\substack{\boldsymbol{\tau}\\ [\mathfrak{S}]}}\right)^{-n}\left(\mathbf{I}_{\substack{\boldsymbol{\tau}\\ [\mathfrak{S}]}}\right)^m &= \left(\mathbf{I}_{\substack{\boldsymbol{\tau}\\ [\mathfrak{S}]}}\right)^{-n}\left(\mathbf{I}_{\substack{\boldsymbol{\tau}\\ [\mathfrak{S}]}}\right)^n\\ \left(\mathbf{I}_{\substack{\boldsymbol{\tau}\\ [\mathfrak{S}]}}\right)^{m-n} &= \mathbf{I}.\end{aligned}$$

In other words, the $(m-n)$-th power of $\mathbf{I}_{\underset{[\mathfrak{S}]}{\boldsymbol{\tau}}}$ is the identity matrix.

$\square$